



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

P – Performance measure
E – Environment
A – Actuators
S – Sensors

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

The environment is represented by `self.width`, `self.height`, `self.food_positions`, `self.walls`, `self.opponents`, and `self.agent_pos`, as these variables define the grid world, obstacles, food locations, opponents, and the agent's position.

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

Multi-Agent. The environment is multi-agent because, in addition to the main agent, there are opponent agents (`self.opponents`) that also move and influence the environment. Their actions can affect the agent by causing collisions and reducing

Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

A percept sequence is the complete history of all percepts (observations) that an agent has received from the beginning until the current moment.

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

Sensors. `get_percept()` represents the Sensors component because it gathers information from the environment and provides it to the agent.

6. (Evaluate) Based on the exact dictionary returned by `get_percept()`, is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

The environment is Partially observable because the agent does not receive complete information about the entire grid. It only receives limited information such as whether it is on food, whether it hit a wall, and opponent positions, rather than

Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

Performance Measure - P

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

Performance measures should evaluate the effect of the agent's actions on the external environment rather than its internal processing. Measuring only internal effort could encourage undesirable behaviour, such as repeatedly performing meaningless actions to gain points without actually achieving the task. Measuring the environment ensures the agent works toward the real objective.

Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

The environment becomes more partially observable because information about the toxic traps exists in the environment but is hidden from the agent's sensors. The agent must make decisions without complete knowledge of the environment.

Step 2.2: Updating the Perception Subsystem (get_percept)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

The new `smells_toxin` sensor provides additional information about dangerous locations. This improves the agent's percept sequence, allowing it to avoid toxic traps and choose safer actions. As a result, the agent behaves more rationally by

Step 2.3: Modifying Action Execution & Visual Rendering (execute_action & draw_grid)

1. **Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

The Vacuum World exploit showed that if a robot vacuum earned points every time it collected dirt, it could repeatedly pick up and drop the same dirt to gain unlimited points. This demonstrated that performance measures must evaluate improvements

Finalize and Lock Lab 01 Analytical Evaluation

